

Classify prime, square and cube numbers

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Classify 97, 144 and 125. Copy or complete the first step.

2. Is 91 prime? Explain.

3. Write the square numbers between 50 and 150.

4. Miro says: Miro says 1 is prime because it has no factor pairs apart from itself. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Classify 97, 144 and 125. Copy or complete the first step.

Answer 97 is prime; 144 is square; 125 is cube.

2. Is 91 prime? Explain.

Answer No, because $91 = 7 \times 13$.

3. Write the square numbers between 50 and 150.

Answer 64, 81, 100, 121 and 144.

4. Miro says: Miro says 1 is prime because it has no factor pairs apart from itself. Is this correct? Explain.

Answer A prime number has exactly two factors. The number 1 has only one factor.

Classify prime, square and cube numbers

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Is 91 prime? Explain.

2. Write the square numbers between 50 and 150.

3. Find a number below 200 that is both a square and a cube.

4. Identify and correct this misconception: Miro says 1 is prime because it has no factor pairs apart from itself.

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. Is 91 prime? Explain.

Answer No, because $91 = 7 \times 13$.

2. Write the square numbers between 50 and 150.

Answer 64, 81, 100, 121 and 144.

3. Find a number below 200 that is both a square and a cube.

Answer 64.

4. Identify and correct this misconception: Miro says 1 is prime because it has no factor pairs apart from itself.

Answer A prime number has exactly two factors. The number 1 has only one factor.

5. Write a complete sentence explaining how you checked one answer.

Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Classify prime, square and cube numbers

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Find a number below 200 that is both a square and a cube.

2. Explain why this reasoning fails: Miro says 1 is prime because it has no factor pairs apart from itself.

3. Create a new example that tests the same idea as: Write the square numbers between 50 and 150.

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. Find a number below 200 that is both a square and a cube.
Answer 64.
2. Explain why this reasoning fails: Miro says 1 is prime because it has no factor pairs apart from itself.
Answer A prime number has exactly two factors. The number 1 has only one factor.
3. Create a new example that tests the same idea as: Write the square numbers between 50 and 150.
Answer Answers vary. The example and solution must preserve the mathematical structure.
4. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
5. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.